

Interleaving is a learning technique in which different topics, types of problems, or skills are mixed together during a single session. Interleaving requires the brain to continually reset and choose the appropriate strategy for the task at hand.

Students often know which formula/lens to use simply because of the unit title. However, cumulative assessments and real-world challenges are not the same. Interleaving can help develop metacognition, as students monitor their own understanding to decide which "tool" from their mental toolbox is required for a specific scenario.

- Interleaving is a "pillar" of effective learning because it manages cognitive load while maximizing the benefits of retrieval.
- Strengthens Differentiation: By seeing different types of problems side-by-side, students learn to differentiate between concepts and identify the "deep structure" of a problem.
- Improves Transfer: When students retrieve knowledge in varied contexts, they become better at applying that knowledge to novel situations and new problems.
- Combats the Forgetting Curve: Interleaving incorporates retrieval, acting as a solution to the Ebbinghaus forgetting curve by forcing active memory reconstruction over time.

Learning Phase

Interleave

a₁ b₁ c₁ b₂ c₂ a₂ c₃ b₃ a₃

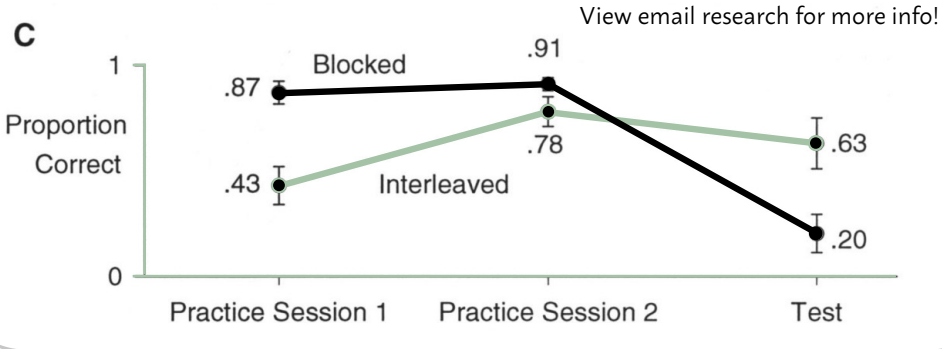
Block

a₁ a₂ a₃ b₁ b₂ b₃ c₁ c₂ c₃

test delay

Test

C



Session	Blocked	Interleaved
Practice Session 1	.87	.43
Practice Session 2	.91	.78
Test	.20	.63

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1. Blocked practice IS effective for learning skills/information short term.

2. Interleaving leads to superior long-term retention and application.

3. Textbooks / traditional course design is dominated by blocked practice, which can lead to an "illusion of mastery."

4. Interleaving is a relatively simple structural switch (no additional time needed, just shifting the order) to boost student retention and the transferability of skills.

It is important to note that interleaving can be a struggle at first

The Challenge: Interleaving feels slower and "messier," often discouraging students.

The Reality: This "desirable difficulty" is what makes learning stick, increasing long-term retention.

The Goal: Using interleaving builds discrimination, the ability to pick the right strategy for real-world problems.

Discipline	Interleaving Example	Why it Works
Arts	Mix prints from three different eras (e.g., Baroque, Impressionist, Pop) and have students identify the movement.	Forces the brain to look for differentiating features rather than just memorizing a single artist's style.
English	Instead of a month of poetry followed by a month of drama, pair a play and a collection of poems thematically.	Requires students to transfer knowledge between different literary forms and structures.
Social Studies	Compare primary sources from three different revolutions (American, French, Haitian) in a single lesson.	Highlights causal nuances and unique variables that are often lost when studying events in isolation.
Math/Science	Give homework sets where problems from the current unit are mixed with problems from two months ago.	Challenges students to choose the correct formula for the problem type, not just use the one they just learned. This interleaves retrieval of old skills with new content, preventing the "learn it and forget it" cycle.
Languages	Conjugate verbs by alternating between three tenses (e.g., Past, Present, Future) in a single response.	Prevents "muscle memory" guessing and forces active memory reconstruction for each specific rule.

Meet with Luke or Jamie



If you have any queries or questions, please do not hesitate to contact Luke or Jamie



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